



Technical Service Bulletin: TSB170096

Released Date: 22-Sep-2017

Fuel Pump Gerotor Over Pressure Protection Feature

Fuel Pump Gerotor Over Pressure Protection Feature

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK19 CM2150 MCRS
- QSK38 CM2150 K106
- QSK38 CM2150 MCRS
- QSK50 CM2150 K107
- QSK50 CM2150 MCRS
- QSK50 DF CM850/CM2150 MCRS
- QSK60 CM2150 MCRS

Issue

Symptom:

- Fault Codes: 2215, 559, 2261, 2262, or 4615 may be present
- Engine stumbles/runs rough, exhibit low power then shuts down
- Engine will **not** restart, check engine light illuminated
- Possible high lubricating oil levels on lubricating oil dipstick

Root Cause:

- Malfunction at fuel pump gerotor caused by outlet restriction. Stage two fuel filter plugging would be an example.

Verification

- Verify presence of Fault Codes: 2215, 559, 2261, 2262, or 4615.
- Verify fuel pump gerotor malfunction caused by increased outlet restriction.

Resolution

- New engine control module (ECM) calibration codes with a fuel pump gerotor overpressure fault code diagnostic strategy are available.

- Install latest ECM calibration code. See corresponding Service Manual. Reference Procedure 019-032 in Section 19. New ECM calibration codes can be identified in Table 1 and Table 2 below.
- New diagnostic is specifically designed to help protect low pressure fuel pump gerotor from stage two fuel filter restriction induced malfunctions. This feature takes into account the effects of engine operating conditions on the fuel supply pressure. The diagnostic actively protects the gear pump by dynamically adjusting operating conditions to keep fuel supply pressure at or below 10 bar.
- Standard troubleshooting procedures for verifying fuel filter restriction across stage two fuel filter head and fuel filter change as required **must** be carried out.

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
QSK19 Parent ECM CM2150	AQ40088.15	FR04510	DO04731
	AQ40089.14	FR04511	DO04732
	AQ40090.15	FR04510	DO04733
	AQ40137.12	FR04549	DO04796
	AQ40138.13	FR04550	DO04797
	AQ40139.09	FR04551	DO04798
	AQ40140.09	FR04553	DO04800
	AQ40142.09	FR04551	DO04804
	AQ40143.09	FR04551	DO04805
	AQ40144.10	FR04551	DO04806
	AQ40145.11	FR04549	DO04807
	AQ40146.13	FR04549	DO04808
	AQ40147.09	FR04549	DO04809
	AQ40148.09	FR04511	DO04810
	AQ40149.09	FR04549	DO04811
	AQ40150.13	FR04555	DO04812
	AQ40152.12	FR04556	DO04815
	AQ40153.09	FR04551	DO04818
	AQ40156.09	FR04549	DO04825
	AQ40157.09	FR04549	DO04827
	AQ40159.13	FR04556	DO04816
	AQ40160.13	FR04556	DO04817
	AQ40163.13	FR04565	DO04834
	AQ40164.11	FR04566	DO04835
	AQ40165.11	FR04567	DO04836
	AQ40166.11	FR04566	DO04840
	AQ40167.11	FR04566	DO04841
	AQ40168.12	FR04564	DO04847
	AQ40169.13	FR04564	DO04833
	AQ40170.10	FR04564	DO04837
	AQ40171.10	FR04564	DO04839
	AQ40172.11	FR04564	DO04849
	AQ40173.12	FR04556	DO04855
	AQ40174.12	FR04556	DO04856
	AQ40175.12	FR04556	DO04857

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
	AQ40178.12	FR04556	DO04865
	AQ40179.12	FR04564	DO04866
	AQ40180.13	FR04556	DO04867
	AQ40181.12	FR04556	DO04868
	AQ40182.12	FR04556	DO04869
	AQ40183.11	FR04510	DO04870
	AQ40184.08	FR04549	DO04875
	AQ40185.08	FR04511	DO04876
	AQ40186.08	FR04549	DO04877
	AQ40187.13	FR04511	DO04878
	AQ40188.09	FR04565	DO04879
	AQ40189.09	FR04567	DO04880
	AQ40190.08	FR04565	DO04881
	AQ40191.08	FR04565	DO04882
	AQ40192.10	FR04566	DO04883
	AQ40193.10	FR04566	DO04884
	AQ40194.09	FR04551	DO04871
	AQ40195.08	FR04551	DO04872
	AQ40196.09	FR04551	DO04874
	AQ40197.10	FR04551	DO04863
	AQ40198.12	FR04556	DO04864
	AQ40200.11	FR04566	DO04886
	AQ40202.10	FR04511	DO04888
	AQ40203.11	FR04556	DO04889
	AQ40204.11	FR04556	DO04890
	AQ40205.10	FR04564	DO04891
	AQ40206.09	FR04510	DO04892
	AQ40207.09	FR04510	DO04893
	AQ40208.08	FR04549	DO04898
	AQ40209.08	FR04549	DO04899
	AQ40210.08	FR04551	DO04900
	AQ40211.08	FR04551	DO04901
	AQ40212.08	FR04551	DO04902
	AQ40213.08	FR04511	DO04903
	AQ40214.08	FR04511	DO04904

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
	AQ40215.09	FR04511	DO04905
	AQ40216.09	FR04511	DO04906
	AQ40217.09	FR04511	DO04907
	AQ40218.09	FR04510	DO04894
	AQ40219.09	FR04510	DO04895
	AQ40220.12	FR04556	DO04897
	AQ40221.08	FR04565	DO04908
	AQ40222.10	FR04566	DO04909
	AQ40223.09	FR04566	DO04910
	AQ40224.10	FR04566	DO04911
	AQ40225.11	FR04564	DO04914
	AQ40226.14	FR04556	DO04917
	AQ40227.10	FR04571	DO04851
	AQ40228.09	FR04549	DO04913
	AQ40229.10	FR04571	DO04915
	AQ40230.09	FR04571	DO04918
	AQ40232.09	FR04549	DO04929
	AQ40233.08	FR04551	DO04948
	AQ40234.10	FR04556	DO04932
	AQ40239.09	FR04551	DO04950
	AQ40240.08	FR04549	DO04957
	AQ40241.09	FR04549	DO04955
	AQ40245.10	FR04556	DO04959
	AQ40247.10	FR04571	DO04965
	AQ40248.09	FR04510	DO04960
	AQ40249.09	FR04511	DO04961
	AQ40250.11	FR04556	DO04962
	AQ40251.09	FR04566	DO04963
	AQ40252.09	FR04567	DO04964
	AQ40253.09	FR04551	DO04971
	AQ40254.08	FR04555	DO04972
	AQ40258.08	FR04550	DO04988
	AQ40259.07	FR04551	DO04989
	AQ40260.08	FR04555	DO04990
	AQ40261.09	FR04571	DO04991

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
	AQ40262.07	FR04571	DO04984
	AQ40263.05	FR04551	DO04985
	AQ40264.05	FR04551	DO04986
	AQ40277.05	FR04522	DO40022
	AQ40278.06	FR04565	DO40023
	AQ40279.07	FR04556	DO40024
	AQ40281.04	FR04549	DO40031
	AQ40284.06	FR04556	DO40042
	AQ40285.06	FR04555	DO40052
	AQ40286.05	FR04551	DO40059
	AQ40287.05	FR04566	DO40047
	AQ40288.05	FR04556	DO40039
	AQ40289.05	FR04564	DO40074
	AQ40290.04	FR04551	DO40065
	AQ40291.04	FR04549	DO40069
	AQ40292.05	FR04565	DO40075
	AQ40293.06	FR04556	DO40076
	AQ40294.04	FR04549	DO40077
	AQ40296.04	FR04551	DO40049
	AQ40297.06	FR04707	DO40071
	AQ40298.07	FR04708	DO40072
	AQ40299.05	FR04555	DO40086
	AQ40301.03	FR04567	DO40105
	AQ40304.01	FR04549	DO40108
	AQ40305.02	FR04522	DO40109
	AQ40306.01	FR04551	DO40110
	AQ40307.00	FR04744	DO40106

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
QSK38 Parent ECM CM2150	AQ60215.19	FR06735	DO06766
	AQ60268.10	FR06779	DO06850
	AQ60269.11	FR06780	DO06851
	AQ60282.11	FR06853	DO06968
	AQ60283.07	FR06854	DO06969
	AQ60284.10	FR06855	DO06970
	AQ60300.12	FR06735	DO06987
	AQ60319.15	FR06735	DO06996
	AQ60339.13	FR06735	DO60063
	AQ60340.14	FR06735	DO60064
	AQ60351.10	FR06735	DO60105
	AQ60393.10	FR06904	DO60110
	AQ60396.09	FR06779	DO60159
	AQ60408.08	FR06842	DO60107
	AQ60409.08	FR06844	DO60109
	AQ60430.07	FR06843	DO60108
	AQ60434.10	FR06735	DO60216
	AQ60435.11	FR06735	DO60235
	AQ60436.10	FR06779	DO60237
	AQ60437.11	FR06780	DO60238
	AQ60459.10	FR06853	DO60372
	AQ60474.07	FR06842	DO60367
	AQ60478.07	FR06853	DO60381
	AQ60505.11	FR06869	DO06984
	AQ60713.07	FR06780	DO60756
	AQ60714.07	FR06855	DO60757
	AQ60722.06	FR06240	DO60765
	AQ60723.05	FR06735	DO60782
	AQ60736.05	FR06869	DO60734
	AQ60737.05	FR06904	DO60712
	AQ60782.00	FR06843	DO60966
	AQ60783.00	FR06844	DO60967
	AQ60784.00	FR06842	DO60968

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
QSK50 Parent ECM CM2150	AQ60217.18	FR06733	DO06765
	AQ60220.18	FR06736	DO06767
	AQ60252.13	FR06781	DO06928
	AQ60287.13	FR06793	DO06853
	AQ60288.13	FR06794	DO06866
	AQ60289.15	FR06795	DO06871
	AQ60292.10	FR06796	DO06977
	AQ60293.10	FR06734	DO06978
	AQ60294.11	FR06746	DO06979
	AQ60296.11	FR06830	DO06976
	AQ60297.10	FR06831	DO06980
	AQ60301.13	FR06746	DO06988
	AQ60309.13	FR06859	DO06981
	AQ60310.13	FR06860	DO06982
	AQ60320.11	FR06792	DO60018
	AQ60338.15	FR06898	DO60081
	AQ60359.14	FR06907	DO60010
	AQ60360.13	FR06908	DO60011
	AQ60361.13	FR06898	DO60012
	AQ60362.13	FR06863	DO60013
	AQ60363.15	FR06864	DO60096
	AQ60364.13	FR06865	DO60097
	AQ60365.14	FR06736	DO60098
	AQ60366.13	FR06890	DO60099
	AQ60367.13	FR06890	DO60123
	AQ60368.13	FR06890	DO60124
	AQ60369.13	FR06888	DO60125
	AQ60370.13	FR06888	DO60126
	AQ60371.13	FR06888	DO60127
	AQ60373.15	FR06888	DO60118
	AQ60374.14	FR06888	DO60119
	AQ60375.14	FR06888	DO60120
	AQ60376.12	FR06746	DO60121
	AQ60377.14	FR06888	DO60122
	AQ60378.14	FR06888	DO60100

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
	AQ60379.14	FR06888	DO60101
	AQ60380.13	FR06888	DO60102
	AQ60381.13	FR06888	DO60104
	AQ60386.15	FR06889	DO60103
	AQ60387.16	FR06889	DO60128
	AQ60388.14	FR06889	DO60129
	AQ60389.13	FR06889	DO60130
	AQ60390.13	FR06889	DO60131
	AQ60391.15	FR06890	DO60133
	AQ60392.11	FR06890	DO60134
	AQ60395.14	FR06890	DO60132
	AQ60397.13	FR06890	DO60157
	AQ60399.14	FR06890	DO60148
	AQ60401.13	FR06890	DO60038
	AQ60402.13	FR06890	DO60135
	AQ60404.14	FR06938	DO60154
	AQ60410.12	FR06905	DO60173
	AQ60416.10	FR06866	DO60195
	AQ60417.11	FR06867	DO60202
	AQ60420.09	FR06889	DO60204
	AQ60421.11	FR06889	DO06870
	AQ60429.11	FR06890	DO60026
	AQ60440.12	FR06795	DO60213
	AQ60441.12	FR06831	DO60191
	AQ60442.10	FR06975	DO60197
	AQ60443.15	FR06890	DO60205
	AQ60444.10	FR06890	DO60215
	AQ60445.10	FR06861	DO60236
	AQ60453.12	FR06863	DO60368
	AQ60454.08	FR06888	DO60369
	AQ60455.10	FR06888	DO60370
	AQ60456.11	FR06889	DO60375
	AQ60458.09	FR06890	DO60371
	AQ60463.09	FR06858	DO60228
	AQ60464.10	FR06890	DO60229

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
	AQ60475.11	FR06890	DO60377
	AQ60476.10	FR06888	DO60383
	AQ60479.08	FR06734	DO60405
	AQ60480.08	FR06790	DO60406
	AQ60481.09	FR06795	DO60407
	AQ60484.07	FR06782	DO60418
	AQ60489.06	FR06974	DO60230
	AQ60490.11	FR06794	DO60408
	AQ60491.08	FR06746	DO60409
	AQ60492.08	FR06858	DO60410
	AQ60493.09	FR06734	DO60411
	AQ60498.08	FR06831	DO60445
	AQ60499.10	FR06781	DO60443
	AQ60500.08	FR06773	DO60446
	AQ60504.07	FR06906	DO60447
	AQ60677.07	FR06888	DO60677
	AQ60690.05	FR06890	DO60629
	AQ60691.08	FR06898	DO60711
	AQ60694.06	FR06890	DO60646
	AQ60695.06	FR06890	DO60737
	AQ60696.06	FR06888	DO60739
	AQ60700.08	FR06888	DO60741
	AQ60701.05	FR06888	DO60747
	AQ60751.03	FR06897	DO60849
	AQ60769.02	FR06736	DO60940
	AQ60770.01	FR06888	DO60939
	AQ60773.01	FR06890	DO60949
	AQ60775.00	FR06860	DO60969
	AQ60776.00	FR06861	DO60970
	AQ60785.00	FR06794	DO60979
	AQ60786.00	FR06901	DO60980

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
QSK60 Parent ECM CM2150	AQ60218.20	FR06734	DO06764
	AQ60221.18	FR06746	DO06792
	AQ60236.17	FR06751	DO06803
	AQ60237.18	FR06751	DO06804
	AQ60238.18	FR06773	DO06839
	AQ60253.18	FR06782	DO06929
	AQ60254.21	FR06790	DO06933
	AQ60255.11	FR06791	DO06931
	AQ60285.09	FR06792	DO06932
	AQ60295.10	FR06829	DO06975
	AQ60302.12	FR06734	DO06868
	AQ60303.17	FR06856	DO06872
	AQ60304.13	FR06857	DO06989
	AQ60305.16	FR06858	DO06869
	AQ60312.12	FR06861	DO60003
	AQ60313.11	FR06974	DO60004
	AQ60314.13	FR06792	DO60002
	AQ60315.11	FR06781	DO60005
	AQ60316.12	FR06792	DO60006
	AQ60317.13	FR06751	DO06852
	AQ60318.10	FR06733	DO06997
	AQ60325.11	FR06792	DO60037
	AQ60326.13	FR06888	DO60020
	AQ60327.15	FR06889	DO60021
	AQ60328.17	FR06890	DO60022
	AQ60329.16	FR06888	DO60023
	AQ60330.13	FR06734	DO60024
	AQ60331.11	FR06895	DO60039
	AQ60335.12	FR06896	DO60048
	AQ60337.12	FR06897	DO06867
	AQ60341.14	FR06899	DO60040
	AQ60342.12	FR06900	DO60042
	AQ60343.12	FR06901	DO60068
	AQ60344.10	FR06895	DO60069
	AQ60345.10	FR06905	DO60070

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60

Engine Model	ECM Code	FR Option	DO Option
	AQ60352.12	FR06906	DO60111
	AQ60398.15	FR06888	DO60147
	AQ60412.08	FR06888	DO60174
	AQ60413.09	FR06890	DO60175
	AQ60415.09	FR06890	DO60198
	AQ60431.11	FR06773	DO60203
	AQ60433.09	FR06938	DO60199
	AQ60446.10	FR06990	DO60233
	AQ60451.08	FR06991	DO60025
	AQ60477.07	FR06906	DO60385
	AQ60482.08	FR06736	DO60227
	AQ60501.08	FR06796	DO60431
	AQ60502.07	FR06792	DO60432
	AQ60503.07	FR06733	DO60433
	AQ60506.07	FR06751	DO60427
	AQ60507.07	FR06791	DO60428
	AQ60508.07	FR06793	DO60429
	AQ60509.10	FR06890	DO60423
	AQ60510.07	FR06794	DO60424
	AQ60511.10	FR06901	DO60425
	AQ60512.09	FR06734	DO60426
	AQ60528.07	FR06751	DO60475
	AQ60530.08	FR06746	DO60478
	AQ60531.08	FR06898	DO60479
	AQ60534.05	FR60088	DO60843
	AQ60716.06	FR06734	DO60761
	AQ60717.06	FR06791	DO60760
	AQ60724.06	FR06908	DO60775
	AQ60748.03	FR06773	DO60827
	AQ60749.04	FR60323	DO60828
	AQ60750.07	FR60323	DO60836
	AQ60759.04	FR06888	DO60884
	AQ60761.05	FR06751	DO60876
	AQ60762.04	FR06773	DO60877
	AQ60763.04	FR06938	DO60878

Table 1, New Primary ECM Codes QSK19, QSK38, QSK50, and QSK60			
Engine Model	ECM Code	FR Option	DO Option
	AQ60777.00	FR68867	DO60971
	AQ60778.00	FR06865	DO60972
	AQ60779.00	FR06864	DO60973
	AQ60780.00	FR06863	DO60974
	AQ60781.00	FR06866	DO60975
	AQ60789.00	FR60429	DO60960
	AQ60790.00	FR60431	DO60962
	AQ60791.00	FR60430	DO60985
	AQ60792.00	FR60444	DO60961
Table 2, New Secondary ECM Codes QSK38, QSK50, and QSK60			
Engine Model	ECM Code	PJ Option	FR Option
QSK38 Child ECM CM2150	AR60190.08	PJ60019	FR06735
	AR60230.05	PJ60098	FR06779
	AR60231.05	PJ60099	FR06780
	AR60250.05	PJ60152	FR06853
	AR60251.05	PJ60153	FR06854
	AR60252.05	PJ60154	FR06855
	AR60327.05	PJ60250	FR06904
	AR60334.04	PJ60247	FR06842
	AR60335.04	PJ60248	FR06843
	AR60336.04	PJ60249	FR06844
	AR60380.04	PJ60362	FR06869
	AR60510.02	PJ60357	FR60240

Table 2, New Secondary ECM Codes QSK38, QSK50, and QSK60

Engine Model	ECM Code	PJ Option	FR Option
QSK50 Child ECM CM2150	AR60193.07	PJ60017	FR06734
	AR60199.07	PJ60020	FR06736
	AR60255.05	PJ60102	FR06782
	AR60257.05	PJ60109	FR06790
	AR60349.04	PJ60117	FR06794
	AR60259.05	PJ60119	FR06795
	AR60265.05	PJ60155	FR06856
	AR60267.05	PJ60157	FR06857
	AR60269.05	PJ60159	FR06858
	AR60271.04	PJ60161	FR06859
	AR60283.04	PJ60163	FR06860
	AR60285.06	PJ60165	FR06861
	AR60321.06	PJ60202	FR06888
	AR60323.06	PJ60204	FR06889
	AR60325.06	PJ60206	FR06890
	AR60328.07	PJ60224	FR06905
	AR60343.04	PJ60295	FR06974
	AR60347.04	PJ60297	FR06975
	AR60354.04	PJ60301	FR06901
	AR60360.04	PJ60312	FR06990
	AR60362.04	PJ60314	FR06991
	AR60194.07	PJ60018	FR06734
	AR60200.07	PJ60021	FR06736
	AR60256.05	PJ60103	FR06782
	AR60258.05	PJ60110	FR06790
	AR60350.04	PJ60118	FR06794
	AR60260.05	PJ60120	FR06795
	AR60266.05	PJ60156	FR06856
	AR60268.05	PJ60158	FR06857
	AR60270.05	PJ60160	FR06858
	AR60272.04	PJ60162	FR06859
	AR60284.04	PJ60164	FR06860
	AR60286.06	PJ60166	FR06861
	AR60322.05	PJ60203	FR06888
	AR60324.05	PJ60205	FR06889

Table 2, New Secondary ECM Codes QSK38, QSK50, and QSK60			
Engine Model	ECM Code	PJ Option	FR Option
	AR60326.05	PJ60207	FR06890
	AR60355.04	PJ60221	FR06901
	AR60329.07	PJ60225	FR06905
	AR60344.04	PJ60296	FR06974
	AR60348.04	PJ60298	FR06975
	AR60361.04	PJ60313	FR06990
	AR60363.04	PJ60315	FR06991

Table 2, New Secondary ECM Codes QSK38, QSK50, and QSK60

Engine Model	ECM Code	PJ Option	FR Option
QSK60 Child ECM CM2150	AR60201.09	PJ60041	FR06746
	AR60202.09	PJ60042	FR06746
	AR60307.06	PJ60111	FR06791
	AR60308.06	PJ60112	FR06791
	AR60275.06	PJ60113	FR06792
	AR60276.06	PJ60114	FR06792
	AR60279.06	PJ60115	FR06793
	AR60280.06	PJ60116	FR06793
	AR60277.06	PJ60121	FR06796
	AR60278.06	PJ60122	FR06796
	AR60293.06	PJ60208	FR06895
	AR60294.06	PJ60209	FR06895
	AR60295.06	PJ60210	FR06896
	AR60296.06	PJ60211	FR06896
	AR60299.06	PJ60214	FR06898
	AR60300.06	PJ60215	FR06898
	AR60358.04	PJ60218	FR06900
	AR60359.04	PJ60219	FR06900
	AR60303.06	PJ60226	FR06906
	AR60304.06	PJ60227	FR06906
	AR60309.06	PJ60228	FR06907
	AR60310.06	PJ60229	FR06907
	AR60311.06	PJ60230	FR06908
	AR60312.06	PJ60231	FR06908
	AR60313.04	PJ60241	FR06863
	AR60314.04	PJ60242	FR06863
	AR60315.04	PJ60243	FR06864
	AR60316.04	PJ60244	FR06864
	AR60317.04	PJ60245	FR06865
	AR60318.04	PJ60246	FR06865
	AR60337.04	PJ60279	FR06866
	AR60338.04	PJ60280	FR06866
	AR60339.04	PJ60281	FR06867
	AR60340.04	PJ60282	FR06867
	AR60195.06	PJ60015	FR06733

Table 2, New Secondary ECM Codes QSK38, QSK50, and QSK60

Engine Model	ECM Code	PJ Option	FR Option
	AR60196.06	PJ60016	FR06733
	AR60210.06	PJ60049	FR06751
	AR60212.05	PJ60092	FR06773
	AR60213.05	PJ60093	FR06773
	AR60211.06	PJ60095	FR06751
	AR60287.04	PJ60100	FR06781
	AR60288.04	PJ60101	FR06781
	AR60297.04	PJ60212	FR06897
	AR60298.04	PJ60213	FR06897
	AR60301.04	PJ60216	FR06899
	AR60302.04	PJ60217	FR06899
	AR60330.04	PJ60251	FR06938
	AR60331.04	PJ60252	FR06938
	AR60405.03	PJ60435	FR60088
	AR60406.03	PJ60436	FR60088
	AR60527.00	PJ60694	FR60323
	AR60528.00	PJ60695	FR60323
	AR60221.04	PJ60143	FR06829
	AR60222.04	PJ60144	FR06829
	AR60253.04	PJ60145	FR06830
	AR60254.04	PJ60146	FR06830
	AR60219.09	PJ60147	FR06831
	AR60220.08	PJ60148	FR06831

Description of Change

New fault code diagnostic is comprised of four fault codes, summarized in Table 3 below.

Table 3, Fault Code Theory of Operation

Fault Code	Latching	Lamp	SPN	FMI	Condition for Activation	Effect	Conditions for Clearing
FC2261	Yes	White/Maintenance	94	15	Fuel supply pressure is above 1000 kPa [145 psi] for more than three seconds.	None	Powering down ECM followed by engine speed above 600 rpm and fuel supply pressure below 1000 kPa [145 psi]
FC2216	Yes	Yellow/Amber	94	16	Fuel supply pressure is above 1100 kPa [160 psi] for more than three seconds.	None	Powering down ECM followed by engine speed above 600 rpm and fuel supply pressure below 1100 kPa [160 psi]

Table 3, Fault Code Theory of Operation

Fault Code	Latching	Lamp	SPN	FMI	Condition for Activation	Effect	Conditions for Clearing
FC4615	Yes (With Speed Derate)	Yellow/Amber	94	0	Fuel supply pressure is above 1200 kPa [174 psi] for more than three seconds.	Ramped speed derate attempts to lower fuel supply pressure to 1000 kPa or below. Speed derate is limited to a minimum target of 1500 rpm.	Powering down ECM followed by engine speed above 600 rpm and fuel supply pressure below 1200 kPa [174 psi]
FC6719	No	Red	94	0	Fuel supply pressure is above 1200 kPa [174 psi] for more than 30 minutes	System actions can differ based on original equipment manufacturer (OEM) red lamp protocol.	Key OFF. Wait for engine shut down. Key ON and wait for one minute. Crank engine.

Additional Fault Code Information:

- Fault code diagnostic **only** runs:
 - If engine is running at idle or higher speeds (engine speed **must** be greater than, or equal to 500 rpm, and fueling **must** be greater than 20 mg per stroke.
 - Engine is warm enough (fuel temperature **must** be greater than, or equal to -10 °C [14 °F])
- Higher severity fault codes do **not** mask lower severity fault codes.

- **Only** FC6719 (red lamp) is cleared at every key-cycle. FC2261, FC2216, and FC4615 are **not** cleared at key-cycle.
- Once fault codes are invoked, **only** way to clear fault codes will be to perform following steps:
 - Change fuel filters
 - Run engine at rated speed for a few minutes.

Document History

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2017-8-31	Module Created

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